

Obama cornered...

...in 4k. VFX studios use Obama's visit to dreamworks to protest their grievances
<http://dgit.in/vfxprotest>



Kindle Fire hdx 8.9

The Kindle Fire HDX 8.9 reviewed with all its pros and cons. Read more if you're thinking of buying one <http://dgit.in/kfire89>

From the labs

successfully alter a physical or a behavioral trait, you cannot just deliver a gene to a group of cells. You'll have to deliver it to each and every cell in the human body. All 100 trillion of them. So the better idea would be to get them while they are young, before the cells have divided into an ungainly amount. More specifically, transferring of a gene into the cells of an early embryo, just after the sperm and the egg meet. If you are in a hurry, you could even transfer a gene into a sperm or an egg cell prior to their rendezvous. This alteration would be permanent throughout the life of the baby.

Are we doing anything with gene therapy?

Well, we do have web-spinning goats. Researchers took the silk-spinning gene of a spider and inserted it into the DNA of a goat in such a way that the animal would make the silk protein in its milk. The protein can then be taken out from the milk and used to create a material with properties comparable to and even surpassing steel, called Biosteel, which is exactly what it sounds like.

But let us take this even further. Theories suggest that the protein responsible for the toughness of our skin- keratin, can be replaced with this tough silk protein, which can supposedly deflect bullets.

Gene therapy has also, in one instance, given us cows which fart and burp less. This is great since cow-gas is considered 20 times more potent than carbon dioxide as a greenhouse gas.

Also, poison cabbage. Scientists are in the process of taking out the gene responsible for programming poison from a

scorpion's tail and inserting it into a cabbage. This has been done with the aim of reducing the use of pesticide, and the poison would be modified to be harmless to humans.

Some South Korean scientists were able to create cats which literally glow (in ultraviolet light though) by inserting, through a virus, genetic instructions for making red fluorescent protein in the cats. While the scientists said they did this as it would help them treat human genetic diseases, we know the only reason why it was done. Glowing kitties.

What about people?

On September 14, 1990, four-year old Ashanti De'Silva was the first successful gene therapy patient. She had lacked a certain enzyme due to which her immune system was lacking. Doctors took white blood cells from her, inserted the normal genes for making the required enzyme into them, and then reinjected it into her.

In another instance, gene therapy through a single injection improved to a slight extent the eyesight of 12 people who were suffering from retinal degeneration caused due to a rare inherited sight disorder, out of whom one kid recovered to almost normal light sensitivity.

There have even been breakthroughs in research of gene-therapy being used for advanced heart failure patients.

If it is so useful, why isn't it in common use yet?

Because there were some awful mess-ups.

In the late 1990's, gene therapy was applied to some children suffering from X-linked Severe Combined Immune Deficiency, or the 'Bubble-boy Syndrome' where the child has an almost non-existent immune system, thus being forced to remain in a sterile environment, which can translate to living in a bubble.

The children were all cured, but two got leukemia, or cancer of the blood. While the leukemia was also subsequently cured with chemotherapy, the



Bubble boy syndrome

fact that the kids even got leukemia raised some serious doubts about the whole process. But this was tame compared to Jesse Gelsinger's fate.

The 18 year old boy suffered from a rare metabolic disorder that prevented his body from getting rid of the toxic product caused by the breakdown of proteins in the body, ammonia. Since this was a single-gene defect, and thus would be easier to cure using gene therapy, Gelsinger enrolled in a clinical trial for gene therapy in 1999. Unfortunately, not long after the researchers had injected the boy with the genes, things went haywire and his ammonia levels went alarmingly high. He suffered brain damage and organ failure within a few days, was in a coma, and eventually his family took him off life support.

Gene therapy, seen then as the new cure on the block, suffered a huge setback due to this tragedy. While certain treatments were definitely practiced, the cases were more isolated and gene therapy struggled to reach the acceptance it had in the early 1990s.

And now gene therapy is back?

Scientists have always been working on gene therapy and its uses on human beings. Unfortunate incidents like the Jesse Gelsinger case just made scientists realize that more caution was required.

And yes, Gene therapy is back. As recently as this year, three children with the bubble boy disease were treated, twenty years after the first treatment of the disease with gene therapy.

Except this time no one got leukemia.



Spider silk in all its glory